Engineering Leadership Skill Set Overlaps

How Staff Engineer, Engineering Manager (EM), Product Manager (PM), Tech Lead Manager (TLM) and Technical Program Manager (TPM) positions overlap in Big Tech and at high-growth startups – and their di

GERGELY OROSZ

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> ## Q: I've noticed quite a bit of similarity between staff engineering roles and engineering management. What is your take on the similarities and differences?

An interesting observation is how many leadership roles in Big Tech and at high-growth startups begin to utilize overlapping skill sets after a while. At these companies, everyone working in engineering or product is technical, meaning they have good understanding of programming and computer-related concepts, and most of them wrote code at some point in their careers.

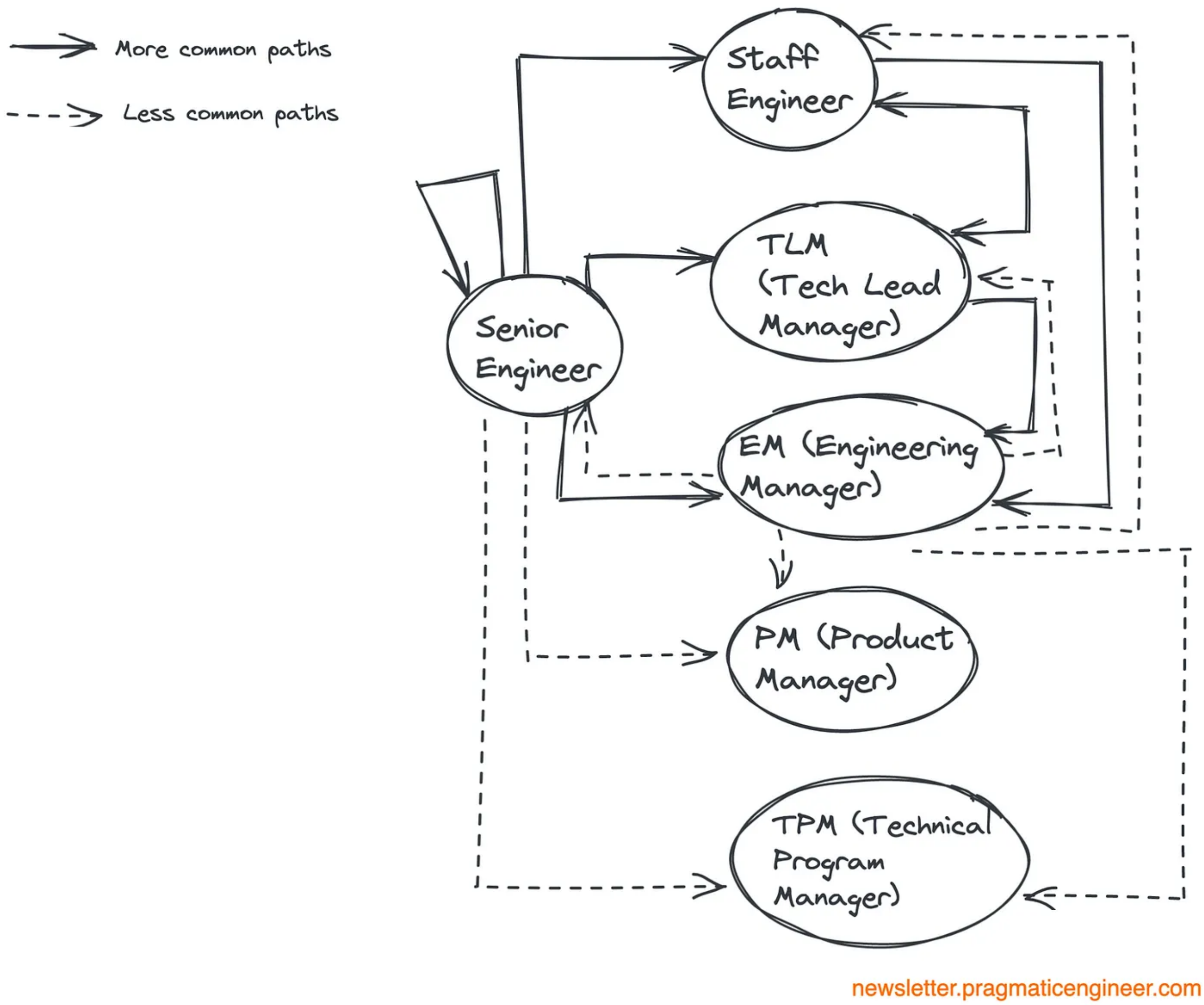
This article covers:

- The most common engineering leadership roles and transitions between them.
- A mental model on how these roles overlap along the strategy & alignment, people management and building software dimensions.
- "Beware": a combination of activities that tend to burn out people.
- Similarities between engineering leadership roles - and areas worth investing in, even if you're not yet in these roles.

Engineering leadership career paths

Which engineering leadership positions are we talking about? We're covering those that typically come after the senior engineer role, and assume a leadership scope that is beyond that role.

These engineering leadership roles are ones that are often progressions from the senior engineer role – given the right opportunity, that is:



More common and less common career paths between engineering leadership roles.

In this article we'll narrow down engineering leadership roles into these five groups:

- Staff Engineer
- Engineering Manager

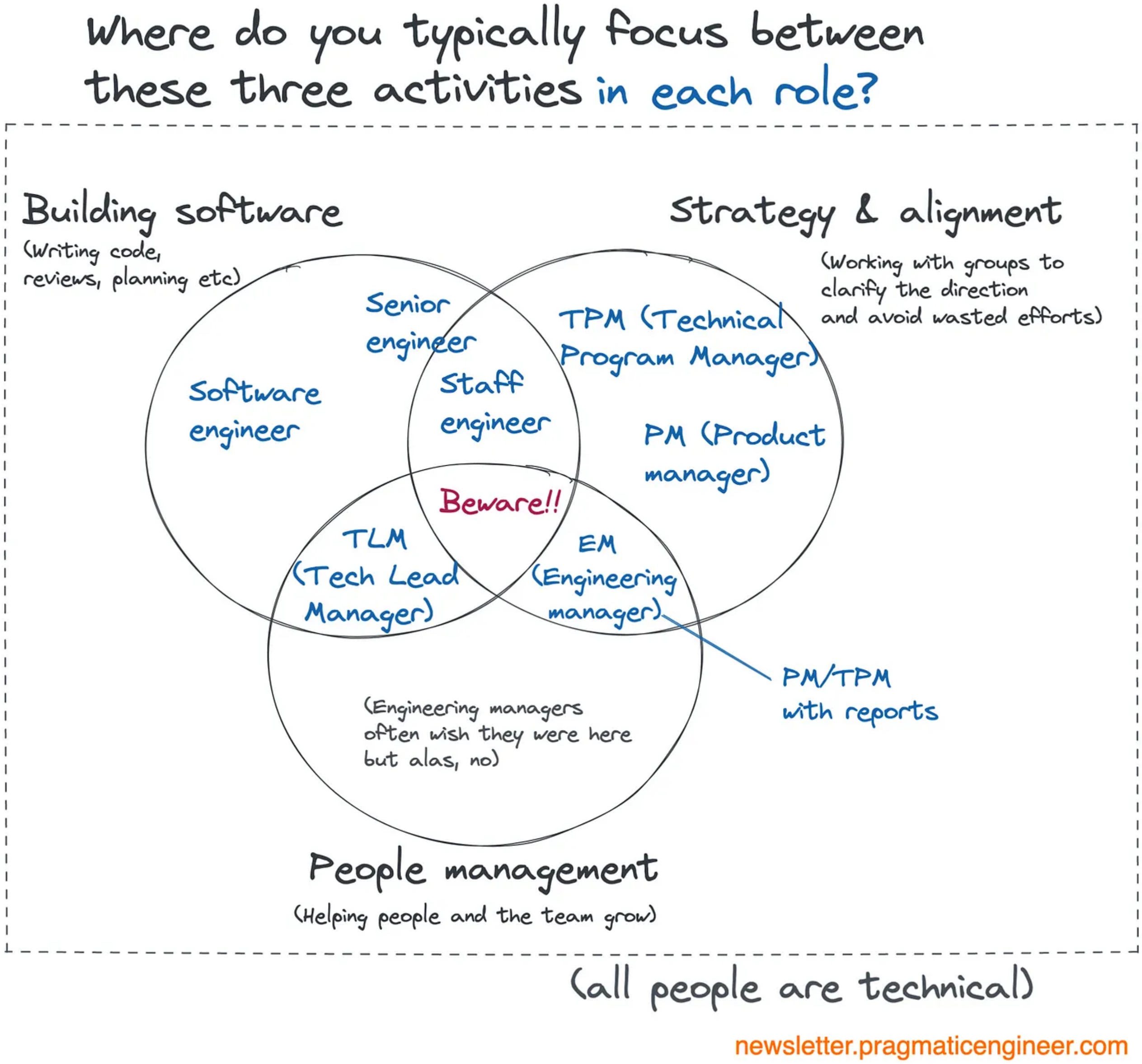
- Tech Lead Manager
- Technical Program Manager
- Product Manager

Strategy & Alignment, People, Building: a Mental Model

A mental model I've started to use is to split day-to-day activities into three buckets, when working in Tech:

- **Strategy and alignment**: clarifying the vision, mission and the strategy of the group. Working with teams and people to align them on the strategy and to avoid wasted efforts. Informing teams, managing dependent teams and collaborating at a team-level.
- **People management:** ensuring the team stays healthy, helping people to grow and the team to execute.
- **Building software**: everything that is directly to do with producing code which powers the products customers use. This goes from planning architecture and code structure, writing code, code reviews, deployments, monitoring of rollouts, and similar activities.

Admittedly, these buckets will not cover all activities engineering leaders potentially do every day. Still, I find this a good enough mental model for visualizing differences between several engineering leadership roles:



Let’s jump into more detail on the roles.

Senior engineer

- **Autonomous**. Typically refers to a fully autonomous engineer who can independently solve almost all complex problems that come up at a team level. This means they can own projects within the team scope end-to-end, unblock themselves and execute reliably.
- **Scope: team level.** Projects they are expected to own and drive, are within the scope of the team. The timeframes of problems they focus on are typically short and medium term, which tend to span one or two projects. Senior engineers are

typically expected to map out dependencies and collaborate with the engineering teams identified as these dependencies. For example, when building a feature to display use data, they might need to rely on the team which owns the user data store.

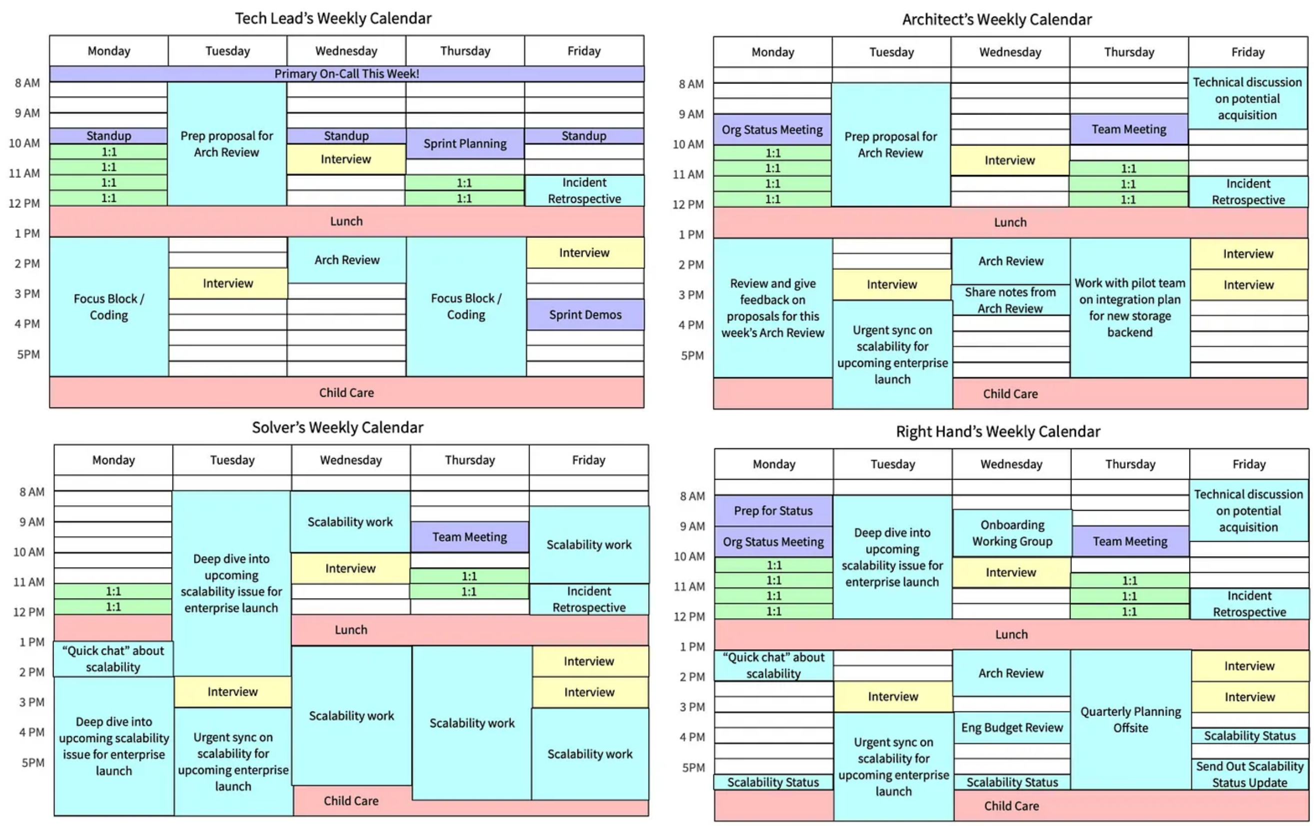
- **Building software: the most amount of time.** Involved in planning, coding, code reviews, engineering practices. The more senior the engineer, the more likely they'll spend more time on key activities that are not coding, but which help the team to build, such as planning, code reviews or tooling.
- **Strategy: little time.** Senior engineers tend to contribute to the roadmap for the team, especially on the technical side. However, they rarely spend time on shaping the engineering strategy of the organization.
- **Alignment: little time**. They are expected to manage stakeholders of projects they are responsible for. These stakeholders are usually other engineers, but they can include business stakeholders, as well. Stakeholder management takes up a smaller fraction of their time though, and they're not expected to be experts at it.
- **Other activities: increasing amount of time**. Mentoring is a frequent expectation; helping other engineers on the team to grow, and being a role model for less experienced engineers on the team. On top of mentoring, senior engineers are often pulled into meetings with other teams, planning, are more frequently asked to interview candidates, or to represent the team at various events. Part of the challenge at the senior engineering level is juggling these additional responsibilities, while still getting things done and unblocking other team members.
- **Most engineers get here**. Most – if not all – software engineers are expected to get to this level, with time, experience, exposure to challenging problems, and opportunities for taking on these challenges.

Staff engineer

The individual contributor career level which is above senior engineer. This position does not involve people management.

The naming of this level can vary between companies and regions. At smaller firms, using the term Principal Engineer or Lead Engineer is more common. I've also heard Tech Lead, Senior Engineer 2 or Chapter Lead all describing a very similar position.

There [are several staff engineer archetypes](), the most common ones have been collected by Will Larson, in his book, [Staff Engineer](). The four most common ones are the Tech Lead, the Architect, the Solver, and the Right Hand. The Staff Archetypes article collects imaginary calendars for each archetype to help visualize the differences between them:



Visualizing weekly calendars for the Tech Lead, the Architect, the Solver and the Right Hand. Source: StaffEng.

How do Staff Engineers fit into the dimensions of People, Building and Strategy/Alignment?

- **Scope: across teams / organization.** Staff-and-above engineers are almost always expected to work across several teams, on projects that need more than one team to deliver them. At Big Tech, staff engineers often report to directors who oversee a whole organization of several teams.

- **Building software: a good chunk of time.** While most staff engineers remain involved in their team's work, they typically spend less time directly working on the codebase due to other responsibilities. They are usually most involved in planning and architecture discussions. At most of Big Tech and at high-growth startups, staff engineers with the 'Tech Lead' or 'Solver' archetypes are still on the oncall rotation, and often champion tooling and process improvements that help with engineering productivity.
- **Strategy: some amount of time.** Staff engineers tend to craft and drive much of the engineering strategy at the organization level. Their involvement depends on how the organization works. However in Big Tech, staff engineers often have an even level of say – and time spent – on engineering strategies, as engineering managers do. For example, at Uber, the revamping of the engineering planning process, originally called [the RFC process](), was driven by a group of staff engineers, and led by one of the two principal engineers at Uber.
- **Alignment: a good chunk of time**. Staff engineers are expected to manage stakeholders for the projects they are responsible for. These stakeholders include other engineering teams, as well as business stakeholders.

Other activities: a good chunk of time. Staff engineers have an increasing amount of additional responsibilities, compared to senior engineers. They are usually pulled into strategic discussions, often serve on promotion panels and are more frequently asked to present their organization at internal and external events. Staff engineers need to be good at both multitasking, as well as carving out focus-time for getting things done.

Engineering manager

A position where the person assuming it is not expected to code day-to-day, but in return manages a larger team of up to around 12 engineers. To make things confusing, companies often call this position Team Lead, Squad Lead or even Tech Lead.

- **Scope: team or organization level.** At the line manager levels, engineering managers are only responsible for their direct team. Starting at the senior engineering manager levels, they start to be responsible for several teams. However, even as a line manager responsible for a single team, in order to be efficient they need to collaborate with peer teams and managers, and to have a good understanding of the lay of the land within their organization and beyond.

- **People management: a good chunk of time.** The main responsibility of an engineering manager is to make their team efficient. To do so, the team needs to be healthy and the EM will need to address any issues that might arise. Engineering managers will spend much of their time on helping the team operate well and people to grow. This will encompass team building and operating activities like 1:1s and roadmapping. It will also encompass hiring, promotions, coaching / mentoring / giving feedback and performance management. See some other activities in the 🔒 (new) engineering manager checklist.
- **Building software: little to no time.** There are two schools of thought on whether engineering managers should be hands-on with coding. For the sake of this split, I'm assuming they don't work on production code and don't inject themselves into the critical path. We'll discuss the engineering manager archetype that does so, in the following section.
- **Strategy: some amount of time.** Engineering managers drive the technical strategy of their team, and are often involved in the organizational-level discussions and decision making.
- **Alignment: some amount of time**. Even with a strong product counterpart and a good product-engineering relationship, engineering managers typically need to spend time aligning with other teams to unblock their engineers. In practice, this often means taking up activities like representing the team in relevant discussions, following up dependencies and taking on most of the burden of the regular planning processes.
- **Other responsibilities: a good chunk of time.** Engineering managers tend to get pulled into a variety of activities. These include hiring for other teams, like recruiting fellow engineering managers, product managers, and sometimes being on the hiring loop for their own future manager. They are frequently pulled into promotion committees, organizing or presenting at org-wide or site-wide events and mentoring people outside of their teams.

The Tech Lead Manager

A position where the person assuming it is expected to both code day-to-day, as well as manage a smaller team of up to five engineers. This position is often called Team Lead or Squad Lead at some places.

Tech Lead Managers *do* spend considerable time building software, unlike engineering managers who don't write code. They also have people management responsibilities, need to manage stakeholders, and have other activities to attend to. On top of all this, Tech Lead Managers are typically less experienced people managers than engineering managers.

So how can Tech Lead Managers deal effectively with more responsibility? They can only manage if their team is smaller in size and scope, meaning they can spend a lot less time on people management, strategy / alignment of stakeholders, than engineering managers would.

Another common best practice is for TLMs to report to an EM who takes care of most of the people management aspects of the job like hiring, career conversations, performance reviews and promotions administration. This frees up the TLM to focus more on the "Building" side, and have some – but not too much – exposure to the People side.

The lack of career progression is a common challenge for Tech Lead Managers. In the short-term, acting in this role helps engineers gain some people management skills. However, unless they switch to engineering management, or return to the individual contributor career ladder, they can typically get stuck in a position that has no career progression.

In my experience, Tech Lead Managers are the most likely to burn out, as they have to juggle the highest number of distinct responsibilities among all engineering leadership positions. They're likely to be pushed to do more people management than agreed upon, they'll often get more involved in building than they intended, and they need to typically align with more stakeholders than they expected. We'll touch on this danger in the "Beware" section.

How do TLMs fit into the dimensions of People, Building and Strategy / Alignment?

- **Scope: team level.** TLMs own the scope of a smaller team.
- **People management: from not much time to a good amount of time.** The amount of people management depends on how involved their manager is in this area, how large their team is, and how much support they get.
- **Building software: some amount of time.** TLMs are still hands-on, and spend a good amount of time planning, coding, reviewing and on other activities directly related to building software.
- **Aligning stakeholders: some time**. Similar to engineering managers, TLMs typically need to manage stakeholders. In an ideal setup, TLMs would have to interface with relatively few such stakeholders.
- **Other responsibilities: a good chunk of time.** Similar to engineering managers, TLMs are often also pulled into a variety of other activities.

Technical Program Managers and Product Managers

The Technical Program Manager is a relatively new position in high-growth organizations and at Big Tech. In the article [What does a Technical Program Manager do] Pat Kua, former CTO at fintech N26, interviewed Meysam Sarabadani, TPM at the company. Meysam summarized the TPM role like this:

> "This role coordinates projects across many teams who all have stakes in a technical project. (...) They need to understand and traverse technical dependencies, work very closely with deeply technical people and at the same time manage the roadmap and often non-technical stakeholders to ensure teams are on track."

I also talked with Senior Technical Project Manager Yixin Zhu, currently a senior manager and TPM at Tempo. I previously worked with Yixin at Uber, where he was responsible for several Uber-wide projects. He summarized the TPM role in an elevator pitch:

> "To simplify where a TPM comes in, let's split up who is responsible for each question:

> **Why** are we building and **what** are we building? – the PM
> **How** will we build it? – the EM
> **When** will we build it and **who** will build it? – the TPM"

How do TPMs fit into the dimensions of People, Building and Strategy / Alignment?

- **Scope: across the organization.** TPMs are always with several teams across the organization, often on complex projects. An example I like to use is a TPM driving General Data Protection Regulation implementation across the organization, which will likely affect all teams.
- **Strategy: a good chunk of time.** TPMs will be heavily involved in the technology and business strategy, being biased towards the technical strategy, and the tactics for realizing this strategy.
- **Aligning stakeholders: a good chunk of time.** TPMs spend a considerable time working with different teams, coordinating large and complex projects and aiming to help all teams progress towards common, organization-level goals.
- **People management: only if they have reports.** Outside of TPMs who have direct reports, most TPMs do not manage people.
- **Other responsibilities: a good chunk of time.** I won't pretend I have all the responsibilities of TPMs covered; they will spend time on more tasks than listed here. *In an article coming early 2022, we'll dive deeper into the TPM role and what engineering managers and engineers can learn from it, and which activities they're doing that are "TPM-like" already.*

Product manager

Many people have described the role of a product manager well. My favorites are Sherif Mansour explaining the role and Lenny Rachitsky answering the question of what is product management in his newsletter. In that article, he writes:

> *"Your job as a PM is to deliver business impact by marshaling the resources of your team to identify and solve the most impactful customer problems. There are three parts to this, each essential:*

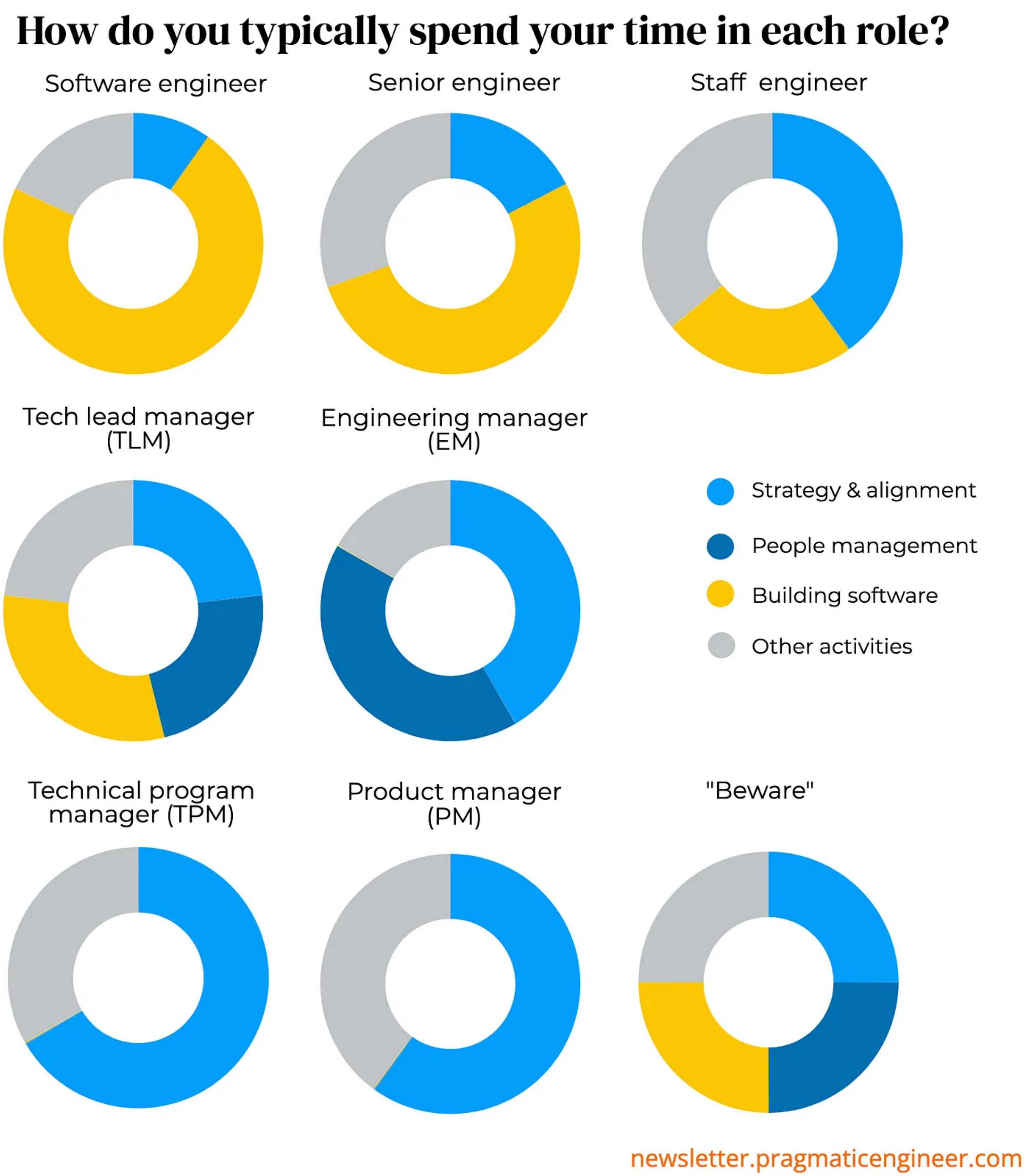
> *1. Deliver business impact (...)*
>
> *2. Marshal the resources of your team (...)*
>
> *3. Identify and solve the most impactful customer problems (...)*
>
> *PMs are the ultimate business lever."*

So how do Product Managers fit into the dimensions of People, Building and Strategy / Alignment? They are similar to TPMs, except their focus is often more on the business side of things, than the technical challenges.

- **Scope: across the organization.** PMs will collaborate with many teams and work on projects that move the needle for the organization, even if they are a partner to only one team.
- **Strategy: a good chunk of time.** Product managers own the "why" and the "what" and drive building out the business strategy for the team. This strategy needs to fit into the organization-wide strategy that they are often involved heavily in.
- **Aligning stakeholders: a good chunk of time.** PMs work with a variety of stakeholders, from design, data science, engineering, and all the way to various business functions. And, of course – most importantly! – their key stakeholder will be customers, potentially both external and internal.
- **People management: only if they have reports.** Whether a product manager has reports or not will depend on the organizational setup and their seniority. Most product managers will not have so many reports to make, although managing these reports takes up the majority of their time.
- **Other responsibilities: a good chunk of time.** Product managers tend to have lots of other responsibilities on their plate, depending on how many stakeholders they work with. Get a sense of these other responsibilities by glancing at the calendar of your product manager, and talking to them about their typical week.

Strategy & Alignment, People, Building, Other: Visualizing these Roles

Let's visualize how each of the above roles would look within the four dimensions of Strategy & alignment, People, Building and Other:



Visualizing how each role might spend their time. Note the role that falls closest to the “Beware” pattern.

While the above splits are a hypothesis – as they can vary greatly depending on the environment – they do paint a picture of how each role will have different focus areas.

There’s one combination we need to talk more about.

“Beware”

I’ve consistently observed that people in roles which force them to mix many different types of activities among these dimensions, tend to struggle more and are more likely to both perform below what is expected of them, and to burn out.

Why is this?

The conflict between the maker’s schedule and the manager’s schedule is a common root cause. Y Combinator cofounder Paul Graham [wrote about this](#):

> “There are two types of schedules, which I'll call the manager's schedule and the maker's schedule. The manager's schedule is for bosses. It's embodied in the traditional appointment book, with each day cut into one hour intervals. You can block off several hours for a single task if you need to, but by default you change what you're doing every hour.
>
> Most powerful people are on the manager's schedule. It's the schedule of command.
>
> But there's another way of using time that's common among people who make things, like programmers and writers. They generally prefer to use time in units of at least half a day. You can't write or program well in hourly units. That's barely enough time to get started.
>
> When you're operating on the maker's schedule, meetings are a disaster. A single meeting can blow a whole afternoon, by breaking it into two pieces each too small to do anything hard in.”

To be efficient at building software, you need to be on the maker's schedule. However, when managing people or aligning teams, you are forced to be on the manager's schedule, attending meetings or being disrupted at any time of day.

In practice, this conflict results either in people not being able to build much software, or those same people being seen as unresponsive to what those on the manager's schedule expect.

People with too many different activities are often inexperienced managers, and this often brings the other common issue to be aware of. Tech Lead Managers are the most likely to fall into the trap of doing all four activities of Strategy & Alignment, People and Building, at once. However, these individuals have typically not managed people before, so they tend to underestimate the time to spend on it. They also have usually not experienced the pressure of doing so many things in parallel, otherwise they would have pushed back on some of these activities.

As a manager, if you spot someone operating in the "beware" zone, try to help them move out of it by doing one of the below. Or something will simply have to give:

- Reduce the "manager's schedule" activities by taking those off their plate.
- Reduce / remove the "maker's schedule" activities by having other people take them over.

Especially with COVID and remote work bringing a new, additional source of stress, beware of setting up people on your team in situations that could lead to them burning out, especially when you can help prevent this.

If you find yourself in the "beware" situation: know that you are at higher risk of performing poorly on all the expectations piled on you. I suggest making a plan to get out of this situation, moving towards either building more software and doing less of the other activities, or doing the opposite by reducing the time you spend building software, in favor of the other activities. Get support from your management chain; they will be better off with someone who does fewer things, than with someone who is on the edge of burning out.

Similarities between engineering leadership roles

Until now, we've focused on the differences in what's expected of various engineering leadership roles. But are there similarities across most engineering leadership roles, from Staff Engineer, all the way to Product Manager?

It turns out there are a few.

1. Technical background. Almost every engineering leader will have a technology background. This means that either they studied technology in school, or have worked in a hands-on individual contributor role related to building software, or have picked up these skills.

This newsletter is written for software engineers and engineering managers. However, if you are not either of these but are looking for advice on becoming more technical without a technical background, Lenny Rachitsky writes about this in his Getting More Technical article. The three steps he outlines in that article are 1. Learn the basics 2. Ask questions 3. Build something.

2. An increasingly long-term view. As you graduate into various engineering leadership roles, your outlook changes from short-term to long-term. You'll have to start to think several steps ahead, understand business cycles, and anticipate changes in the business environment that unfold over greater periods of time.

Getting exposed to a longer-term view tends to be a natural process, if you grow into an engineering leadership role within the same organization. You start to get invited to strategy discussions, to annual or bi-annual planning sessions, and start to make decisions that play out over months, or even a whole year.

3. Deep understanding of the business. To succeed in any of the engineering leadership roles we listed, it's critical that you understand how the business works.

Which areas are money-makers and cash cows? Which are big bets, likely to succeed? Which areas are cost centers, or projects that are at risk of being shut down because they drain too many resources and provide too little return?

If you're a product manager, success in your role depends on understanding where the business is going, and helping it to succeed through your team's work. If you're not a product manager, the best way to build this skill is to [build up a closer relationship with your PM.]()

4. Involved in more streams of work. By being involved in organization-wide projects and working on more complex initiatives, you will naturally work with more teams and stakeholders. You'll also start to be pulled into other initiatives in which your team could be a dependency, now or in the future.

The larger an area you are responsible for, the more of these streams you'll be involved in. As the number of streams increases, it's usually a good idea to pull in other people from your team, if you're a people manager. Doing so both exposes them to other parts of the business, and it helps them grow by letting them represent the team on some of these initiatives.

5. People skills and writing skills are increasingly important. Working with and influencing others becomes important, as does politics. This is a hint to say that you have the tools and credibility to influence larger groups of people in a professional setting. This is especially so with remote work becoming more common and the asynchronous nature of work being more prominent, that writing skill becomes correspondingly more important.

Takeaways

There is an increasing number of options for engineering career paths open to those who are at, or above, the senior engineering level. In this issue, we looked at how these roles compare with each other through the time spent on the activities of Strategy & Alignment, People and Building.

While the roles are different in their expectations and the day-to-day work, they share several similarities. No matter where you are in your career, investing in the five areas below will pay off for you, if and when you get into an engineering leadership role:

1. Technical chops.

2. Taking a longer-term view on projects and initiatives.
3. Deep understanding of the business.
4. Balancing being involved in several workstreams.
5. People skills and writing skills.

I hope this overview provides a useful mental model of thinking about different roles, and helps you decide if and when to take an opportunity which exposes you to a new or different area. And if you find yourself doing too many different types of activities: do be aware!

This is the last longform article of the year. Next Tuesday's article will be a roundup of 2021, before we jump back into the thick of it.

Happy holidays! 🤶 🎄

Thanks to João and Yohan for their input and help with this article.

🤔 How would you rate this week's newsletter?

Amazing • *Great* • *Good* • *OK* • *So-so*

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97 Likes

8 Comments

Write a comment...

Oleksii Asiutin Dec 22, 2021 ♥ Liked by Gergely Orosz

Gergely, the article is amazing, thanks you for writing it!

Where do you think a Staff Engineer should spend "spare time"? Where should Engineering Manager spent this time?

Now I'm working in a big company and the "Spend your time" diagrams look really relevant and correlate with my experience here. But let's consider we have a middle-size company and a Staff Engineer person. It might be the case when there are not much "strategy&alignment" and "other" activities. Of course you do "Building software" activities too. But where do you suggest to spend this, a kind of, spare time? Of course each case is unique, but – diving deep into Building software or spend more time on "Strategy&Alignment", even if you think everything is settled for now (review it again, read relevant books, posts)?

Thanks,

Oleksii

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2 replies by Gergely Orosz and others

Vignesh Chandramohan Writes Vignesh's Substack Dec 22, 2021

Liked by Gergely Orosz

Very well organized post! The part about how staff engineer, EM roles overlap in some aspects were relatable. The way roles work also tend to be very different based on where the project is. Previous newsletter on working with product managers also adds more points to this topic.

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